5 **(a)** Define *magnetic flux density*.

.....

.....

..... [2]

(b) A cyclotron accelerates charged particles to high energies. The simplified diagram in Fig. 5.1 shows the two 'dees' of a cyclotron with a gap in between. There is an electric field at the gap created by a high frequency alternating potential difference. The dashed line shows the path of an accelerated proton. In the shaded region, a uniform magnetic field acts perpendicularly to the plane of the paper.

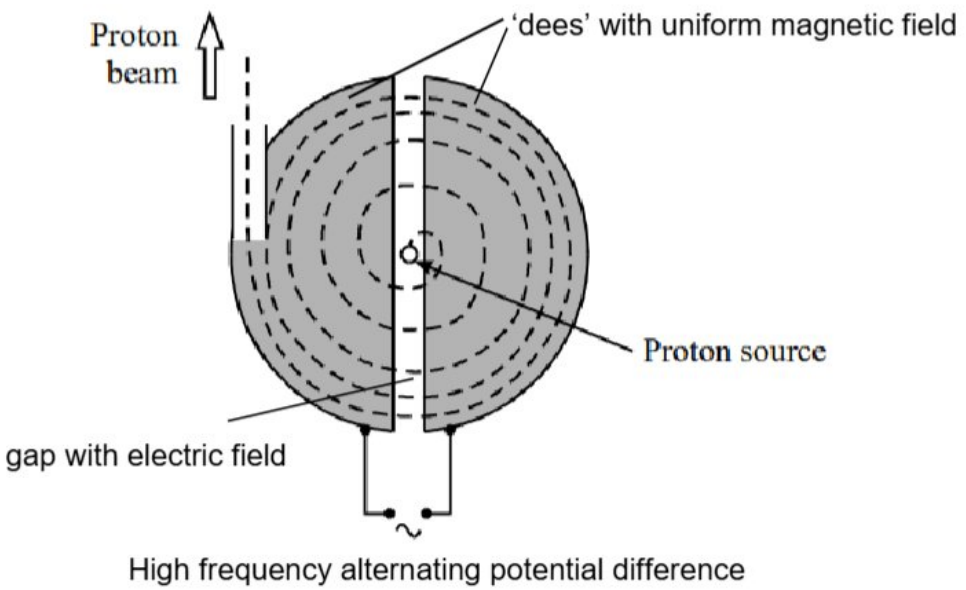


Fig. 5.1

(i) State and explain if the direction of the magnetic field is into the plane or out of the plane of the paper.

.....

.....

.....

..... [2]

(ii) By considering a proton of mass m and charge e moving in a circle of radius r in a uniform magnetic field B, show that the kinetic energy KE can be expressed as

$$KE = \frac{B^2 e^2 r^2}{2m}$$

[2]

(iii) The potential difference across the gap is reversed every time the proton completes half a revolution. Explain how the alternating potential difference increases the kinetic energy of the proton continuously.

.....

.....

.....

.....

..... [2]